Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Improved battery efficiency, reliability, and safety through non-invasive, spatially resolved mapping of currents, losses, and material states			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (x), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (), Cybersecurity ( ), AI (), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Electrics & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Secondary batteries and energy storage	Testing/Inspection/Analysis Techniques	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	The project develops and validates a measurement tool (target technology readiness level 5–6) that localizes current flows, loss mechanisms, and material heterogeneities in battery cells during charge–discharge and electrochemical impedance spectroscopy.			
Rationale for Joint R&D Project	Korea faces intense global battery competition and must improve efficiency, reliability, and safety. ACTRO leads tool industrialization and factory integration;			
Key Words	Advanced battery characterization; production line monitoring; non-destructive evaluation; electrochemical impedance spectroscopy; current mapping; loss localization			

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	- Develop an integrated, non-destructive measurement tool that combines high-frequency three-dimensional magnetic field imaging and lock-in thermography with automated data evaluation. - Achieve high spatial resolution and short measurement cycles suitable for battery production and development workflows. - Detect and rank production failures and design limitations; correlate mappings to performance and safety. - Enable improved battery designs based on spatial mappings			
R&D Target Specifications	Method	Technology innovation	Spatial resolution	Measurement time per cyc
	High-frequency 3D magnetic field imaging	New 3D sensor array up to 50 kHz, lock-in operation and coupling with electrochemical impedance spectroscopy; scalable to larger cells	Sub-centimeter	Less than 1 minute
	Lock-in thermography (with electrochemical impedance spectroscopy)	Smart excitation up to 100 kHz; increased spatial resolution; measurement time reduction	Sub-millimeter	Less than 10 minutes
R&D Plan	**Phase**	**Focus and key tasks**		
	Phase 1: Feasibility and design	Define operating windows, and acceptance criteria; design method coupling and fixtures; prepare instrumented lab cells; establish data quality gates and analysis protocols.		
	Phase 2 Method integration and optimization	Integrate high-frequency magnetic field imaging, lock-in thermography with power systems and electrochemical impedance spectroscopy; automate data acquisition and evaluation; shorten measurement cycles; scale to larger cells.		
	Phase 3 Factory-oriented validation	Deploy at ACTRO for production-like testing; detect and rank failures; correlate spatial maps to performance and safety; iterate tool and workflows.		

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Method development for lock-in thermography and magnetic field imaging; test infrastructure and evaluation routines; data analysis workflows; coordination of international R&D tasks.
Expected R&R of Korean SME Partner	Project lead and principal investigator; tool industrialization and integration with power system; automation, user interface, and factory deployment; site testing and acceptance

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Integrated, non-destructive measurement tool at technology readiness level 5–6 for factory environments, with rapid, high-resolution spatial mapping of currents, losses, and material states. - Earlier detection of production failures and design limitations; reduced safety risk and improved reliability and yield. - Documented improvements in battery lifetime, safety analysis accuracy, and Coulombic efficiency for next-generation cells. - Validated workflows, acceptance criteria, and transfer package enabling deployment across Korean battery manufacturers; strengthened international competitiveness of Korean industry.

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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